

Manufacturing Execution System Analysis Report

Generated: July 11, 2026 at 05:21 PM

Executive Summary

{'total_defects': 53, 'period_days': 7, 'peak_day': 'July 6, 2026', 'peak_defects': 19, 'critical_machines': ['Bat-40', 'Bat-41'], 'elevated_machines': ['Mot-50', 'Mot-51'], 'top_root_cause': 'Inadequate Post-Changeover Verification (87% confidence)', 'overall_risk': 'HIGH - Immediate intervention required'}

Title

Battery Cell Variance Defect Analysis Report

Scope

Maintenance Correlation Analysis

Date Range

July 4-10, 2026

Generated

July 10, 2026

Root Causes

- {'rank': 1, 'cause': 'Inadequate Post-Changeover Verification', 'confidence': '87%', 'defects_linked': 14, 'detail': 'Changeover duration 7.62-8.78 min vs 15-20 min standard. Verification steps being skipped under time pressure.', 'impact': 'HIGH'}
- {'rank': 2, 'cause': 'Chronic PM Deficiency on Bat-40 and Bat-41', 'confidence': '82%', 'defects_linked': 13, 'detail': 'OEE 0.33-0.40 vs 0.85 world-class benchmark. PM avg 7.0 min which is 65% below required duration. Bat-40: 8 defects, Bat-41: 5 defects.', 'impact': 'CRITICAL'}
- {'rank': 3, 'cause': 'QC Station Calibration Drift Post Jul 1 Maintenance', 'confidence': '63%', 'defects_linked': 2, 'detail': 'PM performed at 4.5 min which is 77% below required. 3-day drift onset observed after July 1 maintenance event.', 'impact': 'MODERATE'}

Machine Risk Assessment

- {'machine': 'Bat-40', 'risk_level': 'CRITICAL', 'oee_range': '0.34-0.40', 'defects': 8, 'action_required': 'Unplanned PM within 24 hours', 'pm_deadline': 'July 11, 2026'}
- {'machine': 'Bat-41', 'risk_level': 'CRITICAL', 'oee_range': '0.33-0.37', 'defects': 5, 'action_required': 'PM within 48 hours', 'pm_deadline': 'July 12, 2026'}
- {'machine': 'Mot-51', 'risk_level': 'ELEVATED', 'oee_range': '0.51-0.64', 'defects': 9, 'action_required': 'PM within 72 hours', 'pm_deadline': 'July 13, 2026'}
- {'machine': 'Mot-50', 'risk_level': 'ELEVATED', 'oee_range': '0.47-0.51', 'defects': 10, 'action_required': 'PM within 72 hours', 'pm_deadline': 'July 13, 2026'}

- {'machine': 'QC Station', 'risk_level': 'MODERATE', 'defects': 2, 'action_required': 'Recalibration within 48 hours', 'pm_deadline': 'July 12, 2026'}

Operator Patterns

- {'operator': 'Kim Cannon', 'shift': 'Weekend Night', 'defects': 5, 'finding': 'Structurally insufficient PM durations on Weekend Night shift. Systemic scheduling issue, not individual performance.'}
- {'operator': 'Amanda Barry', 'shift': 'Morning', 'defects': 4, 'finding': 'Mot-51 generating 16 downtime events causing Missing Step defects. Machine reliability driving operator-linked defects.'}

Action Plan

Immediate 0 24H: [{'action': 'Execute Unplanned PM on Bat-40', 'owner': 'Maintenance Lead', 'resources': 'Maintenance technician, PM toolkit, 2-3 hours downtime window', 'success_metric': 'OEE improvement to above 0.65 within 48h post-PM; zero Bat-40 defects for 5 consecutive days', 'risk_if_not_done': 'Machine failure risk HIGH; continued defect generation estimated 8-12 additional defects per week'}, {'action': 'Implement Post-Changeover Verification Checklist (minimum 15 min standard)', 'owner': 'Production Supervisor', 'resources': 'Printed or digital checklist, shift briefing 30 min, QA sign-off requirement', 'success_metric': '100% changeover duration compliance above 15 min; zero changeover-linked defects within 7 days', 'risk_if_not_done': '14 defects per week pattern continues; 87% confidence root cause remains unaddressed'}, {'action': 'QC Station Full Recalibration', 'owner': 'QC Engineer', 'resources': 'Calibration equipment, certified technician, 1-2 hour window', 'success_metric': 'Calibration certificate issued; zero QC drift defects for 14 consecutive days', 'risk_if_not_done': 'Calibration drift worsens; defect detection accuracy degrades; downstream quality escapes possible'}]

Short Term 1 7 Days: [{'action': 'Execute PM on Bat-41 within 48 hours', 'owner': 'Maintenance Lead', 'resources': 'Maintenance technician, scheduled downtime slot, PM toolkit', 'success_metric': 'OEE above 0.65 post-PM; defect count from Bat-41 reduced to zero within 5 days', 'risk_if_not_done': 'CRITICAL escalation; OEE 0.33-0.37 indicates imminent failure risk'}, {'action': 'Execute PM on Mot-50 and Mot-51 within 72 hours', 'owner': 'Maintenance Lead', 'resources': 'Two maintenance technicians, coordinated downtime scheduling', 'success_metric': 'OEE above 0.70 for both machines; Mot-51 downtime events reduced from 16 to below 4 per week', 'risk_if_not_done': 'Amanda Barry shift defects continue; Mot-51 downtime events escalate further'}, {'action': 'Enforce Extended PM Duration Standards — minimum 15 min for all machines', 'owner': 'Maintenance Manager', 'resources': 'Updated PM work orders, supervisor sign-off protocol, duration logging in MES', 'success_metric': '100% of PM events logged above minimum duration threshold; zero sub-standard PM events', 'risk_if_not_done': 'PM deficiency root cause persists; 82% confidence cause remains active'}, {'action': 'Weekend Night Shift PM Protocol Upgrade', 'owner': 'Shift Manager and Maintenance Manager', 'resources': 'Revised shift schedule, dedicated PM time blocks, additional technician coverage if needed', 'success_metric': 'Weekend Night PM durations match weekday standards; Kim Cannon shift defects reduced to zero', 'risk_if_not_done': 'Structural PM gap on Weekend Night continues; recurring defect pattern every weekend'}, {'action': 'Operator Procedure Review — Kim Cannon and Amanda Barry Shifts', 'owner': 'Production Supervisor and QA Lead', 'resources': '1-hour review session per operator, updated SOPs, coaching documentation', 'success_metric': 'Both operators confirm updated procedures; defect contribution drops to zero within 2 weeks', 'risk_if_not_done': 'Operator-linked defect patterns persist; morale impact if systemic issues not addressed'}]

Long Term 1 4 Weeks: [{'action': 'Full PM Schedule Overhaul for Bat-40, Bat-41, Mot-50, Mot-51', 'owner': 'Maintenance Manager and Plant Engineer', 'resources': 'MES scheduling module, maintenance planning software, 2-week planning cycle', 'success_metric': 'All 4 machines achieve

OEE above 0.75 within 4 weeks; PM compliance rate above 95%', 'risk_if_not_done': 'Chronic PM deficiency recurs; OEE remains below 0.50 across critical machines'}, {'action': 'Changeover SOP Revision with Mandatory Verification Gate', 'owner': 'Process Engineer and QA Manager', 'resources': 'SOP authoring, QA review cycle, operator training sessions, MES workflow update', 'success_metric': 'Zero changeover-linked defects for 30 consecutive days; 100% SOP compliance on audits', 'risk_if_not_done': 'Root cause 1 at 87% confidence remains permanently unresolved'}, {'action': 'Maintenance Duration Standards Enforcement via MES', 'owner': 'MES Administrator and Maintenance Manager', 'resources': 'MES configuration, alert rules for sub-standard PM durations, reporting dashboard', 'success_metric': 'MES flags 100% of non-compliant PM events in real time; zero undetected short PMs', 'risk_if_not_done': 'PM duration violations continue undetected; defect recurrence within 30-60 days'}, {'action': 'Battery Cell Variance KPI Dashboard and Weekly Review Cadence', 'owner': 'Plant Manager and QA Lead', 'resources': 'MES reporting module, weekly 30-min review meeting, defect trend dashboard', 'success_metric': 'Battery Cell Variance defects reduced by 80% within 4 weeks; weekly defect count below 10', 'risk_if_not_done': 'No early warning system; next spike goes undetected until defect count is critical']}]

Kpis

- {'metric': 'Battery Cell Variance Defects per Week', 'baseline': 53, 'target': 'Below 10 within 4 weeks', 'owner': 'QA Lead'}
- {'metric': 'OEE — Bat-40', 'baseline': '0.34-0.40', 'target': 'Above 0.75 within 2 weeks', 'owner': 'Maintenance Lead'}
- {'metric': 'OEE — Bat-41', 'baseline': '0.33-0.37', 'target': 'Above 0.75 within 2 weeks', 'owner': 'Maintenance Lead'}
- {'metric': 'OEE — Mot-50', 'baseline': '0.47-0.51', 'target': 'Above 0.70 within 2 weeks', 'owner': 'Maintenance Lead'}
- {'metric': 'OEE — Mot-51', 'baseline': '0.51-0.64', 'target': 'Above 0.70 within 2 weeks', 'owner': 'Maintenance Lead'}
- {'metric': 'Changeover Duration Compliance', 'baseline': '7.62-8.78 min (non-compliant)', 'target': '100% above 15 min within 1 week', 'owner': 'Production Supervisor'}
- {'metric': 'PM Duration Compliance', 'baseline': '7.0 min avg (65% below standard)', 'target': '100% above minimum standard within 1 week', 'owner': 'Maintenance Manager'}
- {'metric': 'Mot-51 Downtime Events per Week', 'baseline': 16, 'target': 'Below 4 within 2 weeks', 'owner': 'Maintenance Lead'}

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